Cloud Server Hosting & Annual Cost Evaluation

A Plain-English Business Guide for Decision Makers: **Render vs. Amazon Web Services (AWS)**

Executive Summary:
Our Attendance and Field Workforce application is currently running on a **Free Hosting Server**. Free servers automatically enter **'Sleep Mode'** whenever no one uses the system for 15 minutes, causing a 60-second delay on the next visit. Upgrading to an **Always-On Server** guarantees the application opens **instantly in under 1 second** for all workers 24/7. This document outlines the simplest, most cost-effective path forward.

1. Why Does the Application Pause or Feel Slow Right Now?

Think of our current free server like a **television that goes into 'Power-Saving Sleep Mode'** when no one presses a button for 15 minutes. When an employee arrives in the morning and tries to punch attendance, the server has to wake up, turn on its engines, and reconnect to the database. This creates a **50 to 90 second delay** before the screen responds. To employees and managers, it feels like the application is frozen or broken.

The Solution: A paid server is **Always-On (running 24 hours a day, 365 days a year)**. Whether it is 8:55 AM or midnight, the application responds **instantly in less than 1 second.**

2. Understanding the Options in Plain Business Terms

- **Option A (Render):** Think of Render like renting a *fully furnished executive apartment*. Everything (electricity, security, maintenance, updates) is managed for you. We flip a single switch, pay a fixed small fee, and it runs reliably with **zero developer maintenance**.
- **Option B (AWS - Amazon Web Services):** Think of AWS like *owning a large commercial building*. It offers infinite scalability and global infrastructure. However, it requires technical upkeep, server management, and careful monitoring to avoid complex billing surprises.
- **What About Our Database (MongoDB)?** Good news: **Our database does NOT need to change.** It is already hosted on the cloud (MongoDB Atlas) and safely holds all employee records, face biometrics, and payslips without extra costs.

3. Annual Cost Breakdown (USD & Indian Rupees)

All calculations assume a standard currency exchange rate of **1 USD = Rs. 84**:

Hosting Plan & Provider	Target Capacity	Monthly Cost	Annual (USD)	Annual (INR)	Performance
Render Free Tier Current Setup	Testing Only (1 - 2 users)	$0 / mo	**$0**	**Rs. 0**	**Sleeps after 15m** (60-90s delay)
Render Starter Plan RECOMMENDED (Phase 1)	Up to 25 Companies (500 - 1,500 workers)	**$7.00 / mo**	**$84.00 / yr**	**Rs. 7,050 / yr**	**100% Always-On** (Instant < 1 sec)
AWS Lightsail Amazon Small Business Cloud	Up to 25 Companies (500 - 2,000 workers)	$5.00 / mo (3 Mo Free)	**$45 - $60 / yr**	**Rs. 3,800 - 5,000**	**100% Always-On** (AWS Mumbai ~20ms)
Render Standard Plan Growth Tier	Up to 50 Companies (2,000 - 5,000 workers)	$25.00 / mo	**$300.00 / yr**	**Rs. 25,200 / yr**	**High Performance** (2GB RAM, fast CPU)
AWS Full Enterprise App Runner / ECS Fargate	100+ Companies (10,000+ workers)	$35 - $75 / mo	**$420 - $900 / yr**	**Rs. 35,000 - 75,000**	**Auto-scaling** (Enterprise Grade)

4. Side-by-Side Feature Comparison (Plain English)

Evaluation Factor	Render (Starter Plan)	AWS (Lightsail Cloud)
Server Sleep / Inactivity	**Never sleeps.** Always on 24 hours a day, 7 days a week.	**Never sleeps.** Always on 24 hours a day, 7 days a week.
Speed & Employee Response	Instant response (< 1 sec for camera, login, reports).	Instant response (< 1 sec from AWS Mumbai data center).
Setup & Downtime	**1-Click Upgrade (Zero downtime).** Ready in 60 seconds.	Requires approx. 1 to 2 hours of server configuration.
Server Maintenance	**100% Fully Managed.** Render handles all OS security updates.	Requires manual Linux administration and security monitoring.
Predictable Billing	**Fixed, predictable $7/mo (Rs. 580/mo).** Zero hidden fees.	Fixed on Lightsail ($5/mo), but variable on storage and bandwidth.
Customer Capacity	Easily supports up to 25 companies and 1,500 active employees.	Can scale up to tens of thousands of users across India.
Annual Investment	**$84 / year (approx. Rs. 7,050 / year)**	**$45 - $60 / year (approx. Rs. 3,800 - 5,000 / year)**

5. Which Plan is Best for You Right Now?

Our Professional Recommendation: Start with the Render Starter Plan ($7/mo | approx. Rs. 580/mo | Rs. 7,050/yr)

Why this is the smartest decision for your business today:

1. **Immediately Eliminates the Problem:** Solves the 15-minute sleep issue 100%. Employees and managers will experience instant punch-ins and fast page loading.
2. **Zero Transition Risk:** There is zero code rewrite, no downtime, and no risk of database corruption or payroll discrepancies.
3. **Lowest Total Cost of Ownership:** AWS seems cheaper on paper ($5/mo vs $7/mo), but AWS requires developer hours to manage Linux patches and firewalls. Render has zero management cost.
4. **Clear Growth Upgrade:** When your customer base grows past 25-30 companies (over 2,000 employees), we can transition the system to AWS seamlessly in an afternoon.

6. Simple Action Plan (Next Steps)

- **Step 1 (Immediate & Free):** We have activated an automatic keep-alive ping on our server to keep the current free plan awake during office hours.
- **Step 2 (Recommended Client Approval):** Approve the **Render Starter Plan ($7/month | approx. Rs. 580/mo)** to permanently eliminate cold starts, double the server memory, and deliver a smooth experience for all users.
- **Step 3 (Future Scale):** Re-evaluate AWS when your customer workforce crosses 2,000 active employees.

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Commercial Approval (Client):

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